

SUPPLY CHAIN PERFORMANCE ANALYSIS

Project Overview

About the Project

This project focuses on analyzing supply chain performance using sales, inventory, supplier, warehouse and profitability data.

The objective is to understand business performance, identify profitable products, evaluate inventory and replenishment risks, compare supplier and warehouse performance, and generate practical business recommendations.

Business Problem

Supply chain businesses need to maintain the right balance between sales performance, inventory availability, supplier efficiency and profitability.

This project uses data analysis and visualization to answer important business questions such as:

- How much revenue and profit is generated?
- Which products contribute the most revenue and gross profit?
- Which warehouses have higher replenishment risk?
- How can inventory and replenishment performance be improved?
- Which suppliers and regions contribute the most revenue?
- Where can the business improve operational efficiency?

Dataset Overview

- **Records:** 91,250
- **Time Period:** January 1, 2024 to December 30, 2024
- **Products:** 50 SKUs
- **Warehouses:** 5
- **Suppliers:** 10
- **Regions:** 4
- **Main Data Areas:** Sales, inventory, suppliers, warehouses and profitability

Tools and Technologies

- **Python:** Data cleaning, preprocessing and exploratory data analysis
- **SQL:** Business analysis and performance queries using MySQL
- **Power BI:** Interactive dashboard and KPI visualization

Project Workflow

Raw Dataset → Data Cleaning → Exploratory Data Analysis → SQL Analysis → Power BI Dashboard → Business Insights and Recommendations

Project Outcome

The final outcome is an interactive Supply Chain Performance Dashboard that helps users monitor revenue, units sold, gross margin, gross profit, product performance, regional contribution and warehouse replenishment risk.

DATA PREPARATION & METHODOLOGY

Data Preparation

The raw supply chain dataset was prepared using Python to improve data quality and reliability before analysis.

Data Cleaning Steps

- Loaded the raw dataset using Python.
- Reviewed columns, data types and data quality.
- Checked missing values and duplicate records.
- Standardized column names and data formats.
- Verified numerical, date and categorical fields.
- Exported the cleaned dataset for SQL and Power BI analysis.

Exploratory Data Analysis

Exploratory Data Analysis was performed to understand important business variables.

The analysis focused on:

- Revenue and gross profit distribution
- Monthly revenue trends
- SKU-level sales and profitability
- Warehouse replenishment risk
- Supplier and regional performance

SQL Analysis

The cleaned dataset was imported into MySQL for structured business analysis.

SQL queries were created to evaluate:

- Overall business performance
- Monthly revenue and SKU profitability
- Inventory and replenishment risk
- Promotion performance and management priorities
- Supplier, warehouse and regional performance
- Management priorities

Power BI Dashboard Development

Power BI was used to create an interactive dashboard for monitoring supply chain performance.

The dashboard includes:

- Total Revenue
- Total Units Sold
- Gross Margin %
- Gross Profit
- Revenue by Month
- Top 10 SKUs by Revenue and Gross Profit
- Revenue versus Gross Profit by SKU
- Revenue by Region
- Replenishment Risk Rate by Warehouse
- Month and Region slicers

OVERALL BUSINESS PERFORMANCE

Executive Summary

The Supply Chain Performance Dashboard provides a consolidated view of the company's financial and operational performance. The analysis focuses on revenue, units sold, gross profit and gross margin.

Key Performance Indicators

KPI	Value
Total Revenue	33.43 M
Total Units Sold	2M (Approximately)
Gross Profit	11.09 M
Gross Margin	33.17%

These KPIs provide a consolidated view of the company's financial performance. Revenue and gross profit should be evaluated together to understand both sales volume and profitability.

Key Observations

- Revenue and gross profit provide a clear overview of the company's overall financial performance.
- Gross margin helps evaluate how efficiently revenue is converted into profit.
- Revenue performance should be analyzed together with gross profit to identify profitable business areas.
- Monthly revenue analysis helps identify changes in sales performance over time.
- Product-level analysis is required to identify the highest-performing SKUs.
- Regional and supplier analysis helps understand the major contributors to revenue.
- Warehouse replenishment risk should be monitored to support smooth operations and product availability.

Management Focus

Based on the dashboard, management should focus on:

- Monitoring revenue, gross profit and gross margin together.
- Identifying products with high revenue and strong profitability.
- Reviewing products with high revenue but comparatively lower profit.
- Tracking monthly revenue trends for sales planning and performance monitoring.
- Monitoring warehouse replenishment risk.
- Comparing supplier and regional contribution.
- Improving inventory planning using inventory and replenishment metrics.

Business Value

The dashboard provides management with a single view of important supply chain KPIs. It supports performance monitoring, product prioritization, inventory planning and data-driven decision-making.

PRODUCT, SALES & REGIONAL PERFORMANCE

Monthly Revenue Performance

The monthly revenue chart shows changes in sales performance throughout the year. Revenue increased during the early months, reached a high point around March, and declined during the middle of the year. Revenue improved again toward the end of the year, showing that sales performance changed across different months.

Key Observations

- Revenue was strongest during the early part of the year.
- A decline was observed during the middle months.
- Revenue started recovering toward the end of the year.
- Monthly trends can help management plan inventory, sales activities and future demand.
- Further investigation is required to understand the reasons behind monthly fluctuations.

Product Performance

The dashboard compares the Top 10 SKUs by Revenue and Top 10 SKUs by Gross Profit. The analysis helps identify products that contribute significantly to sales and profitability.

Key Observations

- A small group of SKUs contributes a major share of revenue.
- Top revenue-generating products should be monitored for continued availability.
- High-revenue products may not always be the most profitable products.
- Top gross-profit SKUs should be prioritized for profitability-focused decisions.
- Products with high sales but lower profit margins require further review.
- Product-level performance can support pricing, promotion and inventory decisions.

Revenue versus Gross Profit

The scatter plot compares revenue and gross profit for individual SKUs.

The chart shows a generally positive relationship between revenue and gross profit. Products generating higher revenue also tend to generate higher gross profit, although the relationship differs across SKUs.

This comparison helps identify:

- High-revenue and high-profit products
- Low-revenue and low-profit products
- Products with strong sales but comparatively lower profit
- Products that may require pricing or cost improvement

Regional Performance

The regional revenue chart compares the contribution of North, East, South and West regions.

The revenue contribution is relatively balanced across the four regions, with no single region dominating the overall business.

Business Implications

- Regional performance should be monitored regularly.
- Sales strategies can be adjusted according to regional demand.
- Inventory should be planned according to regional requirements.
- Balanced regional contribution reduces dependence on one market.
- Further analysis can identify region-wise product preferences and growth opportunities.

Overall Conclusion

Product, monthly and regional analysis provides a clearer understanding of where revenue and profit are generated. These insights can support product prioritization, sales planning, regional strategy and inventory decisions.

INVENTORY, REPLENISHMENT & SUPPLIER ANALYSIS

Inventory and Replenishment Risk

The dashboard compares replenishment risk across five warehouses. The replenishment risk rate is relatively close across the warehouses, indicating that inventory risk exists across the supply chain network rather than being concentrated in only one location.

Key Observations

- Replenishment risk rates are close across the warehouses.
- WH_1 shows the lowest replenishment risk among the displayed warehouses.
- WH_5 shows the highest replenishment risk among the displayed warehouses.
- The difference between warehouses is small but should still be monitored.
- Regular inventory reviews can help prevent stock shortages.
- Replenishment decisions should consider stock availability, inventory levels and operational requirements.

Inventory Planning Implications

- Inventory levels should be reviewed regularly.
- Warehouse replenishment risk should be monitored.
- Products with higher inventory requirements should receive planning attention.
- Stock availability should be considered before replenishment decisions.
- Warehouse and inventory metrics can support purchasing and allocation decisions.

Supplier Performance

Supplier-level analysis was performed to understand revenue contribution across suppliers.

The supplier analysis helps identify major suppliers and supports better supplier management.

Key Observations

- Supplier contribution can differ across the supply chain.
- High-contribution suppliers should be monitored for reliability and continuity.
- Supplier performance should be evaluated along with cost, quality and delivery.
- Dependence on a limited number of suppliers may create operational risk.
- Supplier performance data can support negotiation and sourcing decisions.

Business Value

Inventory, replenishment and supplier analysis helps management improve product availability, purchasing decisions and operational efficiency.

Combining warehouse and supplier information can support a more reliable and efficient supply chain.

KEY INSIGHTS, RECOMMENDATIONS & CONCLUSION

Key Insights

The analysis of supply chain data produced the following important insights:

- Total revenue was approximately **33.43M**.
- Total gross profit was approximately **11.09M**.
- Overall gross margin was **33.17%**.
- Revenue performance changed across different months.
- A small group of SKUs contributed significantly to revenue and gross profit.
- Higher revenue does not always guarantee higher profitability.
- Regional revenue contribution was relatively balanced.
- Replenishment risk was present across all five warehouses.

Supplier and warehouse analysis can support better operational planning.

Business Recommendations

1. Monitor Revenue and Profit Together

Management should track revenue, gross profit and gross margin together instead of focusing only on sales growth.

2. Prioritize High-Performing SKUs

High-revenue and high-profit products should receive priority in inventory planning and product availability decisions.

3. Review Low-Margin Products

Products with high revenue but comparatively lower profit should be reviewed for pricing, cost and promotion effectiveness.

4. Improve Inventory Planning

Warehouse replenishment risk should be monitored regularly to reduce stockout risk and maintain product availability.

5. Use Data for Inventory Planning

Inventory and replenishment metrics should be reviewed regularly to support purchasing, stock allocation and warehouse replenishment decisions.

6. Monitor Supplier Contribution

Supplier performance should be reviewed based on revenue contribution, reliability, cost and delivery performance.

7. Review Monthly Trends

Monthly revenue trends should be monitored to identify changes in demand and support sales planning.

Conclusion

This project transformed raw supply chain data into meaningful business insights using Python, SQL and Power BI.

Python supported data cleaning and exploratory analysis. SQL helped answer structured business questions, while Power BI provided an interactive dashboard for monitoring key performance indicators.

The analysis demonstrates how data analytics can support revenue monitoring, product prioritization, inventory planning, supplier evaluation and business decision-making.

Final Outcome

The completed dashboard provides management with a consolidated view of supply chain performance and highlights areas where further operational improvement can be made.

Data-driven analysis can help the business improve visibility, reduce operational risk and make better supply chain decisions.